

Zhuoya Shi

Curriculum Vitae

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Education

- 2024 **Ph. D.**, *New York University, NYC*, Civil Engineering
2017 **M. Sc.**, *University of Southern California, Los Angeles*, Construction Engineering
2014 **B. S.**, *Chang An University, Xi'an*, Construction Cost

Career Summary

Civil Engineering Ph.D. from New York University, minored in Transportation Engineering, specializing in quantitative modeling and urban data analysis. Developed BIM-based automation algorithms (82% precision, 100% recall) and NLP-based information extraction (93% accuracy) for urban systems. Conducted geospatial clustering and interdisciplinary research. Experienced in implementing Large Language Models (LLMs). Strong publication record. Seeks to apply analytical skills to land-use and transport interaction modeling.

Research Experience

- 2017–2024 **Applied Research Engineer**, New York University, Brooklyn, NY
- Led focus groups with 10+ façade consulting teams & city agencies to define BIM requirements.
 - Developed automation algorithm with Python to generate BIM-based customized inspection checklists (precision 82%, recall 100%).
 - Designed BIM-based integrated visualization, improved inspectors' façade condition assessment accuracy (15%), and efficiency (47.6%).
 - Achieved 93% accuracy in natural language processing-based information extraction (bi-LSTM-CRF) from large-scale façade inspection data (precision 95%, recall 96%).
 - Conducted geospatial clustering analysis on NYC buildings, identified 4 neighborhoods with 9 unique defect patterns, providing data-driven insights for city-wide guidance.
 - Leveraged AR to quantify design features' impact on humans through hypothesis testing (t-test), identifying 9 impactful features.
 - Collected participants' body area sensor data with a virtual reality setting to quantify human experiences.
- 2016–2017 **Graduate Research Assistant**, University of Southern California, Los Angeles, CA
- Captured and processed 3D point cloud data from terrestrial laser scanners and UAV imagery.
 - Applied industry-standard software (Point Cloud Library, Recap, CloudCompare) to generate and analyze 3D point clouds for existing building conditions.
 - Investigated and developed methods for integrating scan data into BIM environments for modeling and analysis purposes.

Publications (6 journal articles, 5 conference papers, citations:177, h-index:5 according to Google Scholar)

Refereed Journal Articles

- [J-6]  ²⁰²⁵ **Zhuoya Shi** and Semiha Ergan. "Spatial Clustering and NLP-Based Analysis of Defect Patterns in Urban Façades: Implementation on NYC Buildings." *Cities*, 166, 106182.
- [J-5]  ²⁰²⁵ **Zhuoya Shi**, Keundeok Park, and Semiha Ergan. "Towards a comprehensive and consistent urban façade inspection process: Data-driven analysis of façade inspection reports for capturing inspection requirements." *Journal of Construction Engineering and Management* (*accepted*).
- [J-4]  ²⁰²⁵ **Zhuoya Shi** and Semiha Ergan. "Transforming Urban Façade Condition Assessments with Semantic Data Visualizations and 3D Spatial Layouts from BIMs." *Buildings*, 15(3), 458.
- [J-3]  ²⁰¹⁸ Semiha Ergan, **Zhuoya Shi**, and Xinran Yu. "Towards quantifying human experience in the built environment: A crowdsourcing-based experiment to identify influential architectural design features." *Journal of Building Engineering*, 20, 51-59.
- [J-2]  ²⁰¹⁸ Meida Chen, Eyuphan Koc, **Zhuoya Shi**, and Lucio Soibelman. "Proactive 2D model-based scan planning for existing buildings." *Automation in Construction*, 93, 165-177.
- [J-1]  ²⁰¹⁸ **Zhuoya Shi**, and Semiha Ergan. "Leveraging point cloud data for detecting building façade deteriorations caused by neighboring construction." *Tamap Journal of Engineering*, 2018.

Under-review

- [J-7]  ²⁰²⁵ **Zhuoya Shi** and Semiha Ergan. "A Model-Based and Data-Driven Approach for Comprehensive and Consistent Inspection of Urban Façades. " *Advanced Engineering Informatics Conference Proceedings*
- [C-5]  ²⁰²⁴ **Zhuoya Shi** and Semiha Ergan. "A model-based approach for flexible façade inspection data visualization of urban buildings." *i3CE 2024*, July 28-31, Pittsburgh, PA.
- [C-4]  ²⁰²³ **Zhuoya Shi** and Semiha Ergan. "Model-based Checklist Generation for Façade Safety Inspection Guidance" *i3CE*, June 25-28, Corvallis, OR.
- [C-3]  ²⁰²¹ **Zhuoya Shi** and Semiha Ergan. "An ontology towards BIM-based guidance of building façade maintenance." *Proceedings of the 38th ISARC*, November 2-3, Dubai, UAE.
- [C-2]  ²⁰²⁰ **Zhuoya Shi**, Keundeok Park, and Semiha Ergan. "Towards a comprehensive façade inspection process: An NLP-based analysis of historical façade inspection reports for knowledge discovery" *Proceedings of the 37th ISARC*, October 27-28, Kitakyushu, Japan.
- [C-1]  ²⁰²⁰ **Zhuoya Shi**, and Semiha Ergan. "Towards Point Cloud and Model-Based Urban Façade Inspection: Challenges in the Urban Façade Inspection Process." *Construction Research Congress 2020: Safety, Workforce, and Education*, March 8-10, Tempe, AZ.

Awards and Honors

- 2017–2022 **School of Engineering (SoE) Fellowship**, *New York University*
- 2020 **Provost TA's Scholarship**, *New York University*
- 2018 **Institute of Design & Construction Innovation Hub (IDC) Foundation Scholarship**, *New York University*
- 2011–2014 **First-class Scholarship**, *Changán University*
- 2013 **Merit Student Cadre of Changán University**, *Chang An University*

2011 **Summer Social Practice Advanced Individual**, *Chang An University*

Consulting Experience

- 2022–2023 **VDC Engineer**, *DeSimone Consulting Engineers*, NYC, NY
- Designed a database for reports management, achieving 11% time savings.
 - Led pilot studies and evaluated 6 software tools for inspection practice digital transformation.
 - Conducted basic material take-offs with Auto CAD drawings and organized project documents.
 - Inspected 4 diverse construction projects, providing updates to stakeholders.

Teaching

- 2018–2019 **CE-UY 3513 Construction Scheduling**, *Department of Civil and Urban Engineering*, NYU Tandon, Brooklyn, NY
- Co-developed course curriculum.
 - Designed comprehensive lab materials including project examples and step-by-step tutorials for Primavera P6 and Microsoft Project.
 - Instructed on Critical Path Method (CPM) application, including sequencing, cost allocation, resource constraints, and facilitated hands-on learning for project planning through control.
- 2018–2022 **CEGY 8383 Building Information Modeling (BIM) and Its Application in AEC/FM**, *Department of Civil and Urban Engineering*, NYU Tandon, Brooklyn, NY
- Developed lab tutorials and model examples for architectural and structural modeling with Revit, clash detection with Navisworks, compliance checking with Solibri, point cloud data processing with Autodesk Recap, and 4D simulation with Synchro.
 - Led labs on point cloud capture using 3D laser scanner.
- Fall 2020 **CEUY 3543 Building Information Modeling**, *Department of Civil and Urban Engineering*, NYU Tandon, Brooklyn, NY
- Taught lab sessions on fundamental BIM principles, including model creation, basic clash detection, quantity takeoff.
 - Designed and prepared term project models and materials.
- Spring 2017 **CE 557 Advanced Building Estimating**, *Department of Civil and Environmental Engineering*, USC Viterbi, Los Angeles, CA
- Evaluated student assignments and provided feedback on advanced cost estimating principles and methodologies.

Mentorship and Student Development

- 2018–2021 **ARISE Program**, NYU, Brooklyn, NY
- Guided 4 minority high school students through research projects.
- 2018–2021 **Tandon Undergraduates Summer Research Program**, NYU, Brooklyn, NY
- Led Python and Machine Learning boot-camps with hands-on labs for undergraduate students.
 - Held brainstorming workshops to troubleshoot research bottlenecks and track student progress.
- 2016 **Summer Program's Discover Engineering**, USC, Los Angeles, CA
- Led engineering labs across multiple disciplines for 20+ high school students.

Technical skills

BIM Software AutoCAD | Revit | Navisworks | BIM360 | Primavera P6 | Solibri | Synchro 4D | Dynamo | Recap | Green Building Studio | CloudCompare | Pix4D | FARO SCENE.

Programming Languages	Proficient in Python SQL R; Familiar with Java JavaScript HTML CSS
Libraries and Tools	TensorFlow Keras PyTorch Scikit-learn NLTK OpenCV Pandas NumPy GeoPandas Selenium Matplotlib Seaborn Plotly Git Tableau AWS Microsoft Power BI JupyterLab Google Colab.

Professional Affiliations And Service

Professional Affiliation/Membership

2017-Present **American Society of Civil Engineers (ASCE)**, *Member*

2017-Present **Structural Engineers Association of New York (SEAoNY)**, *Member*

Reviewer

2024-Present Automation in Construction (AUTCON)

2024-Present Advanced Engineering Informatics

2024-Present Applied Energy

2024-Present Buildings

2024-Present Building and Environment

2024-Present Energy and Buildings

2024-Present Safety Science

2018-Present Journal of Building Engineering

2018-Present Journal of Construction Engineering and Management

2024 International Symposium on Automation and Robotics in Construction (ISARC)

2023 ASCE International Conference on Computing in Civil Engineering (i3CE)

2023 International Symposium on Automation and Robotics in Construction (ISARC)

2020 European Group for Intelligent Computing in Engineering

2020 ASCE Construction Research Congress (CRC)